

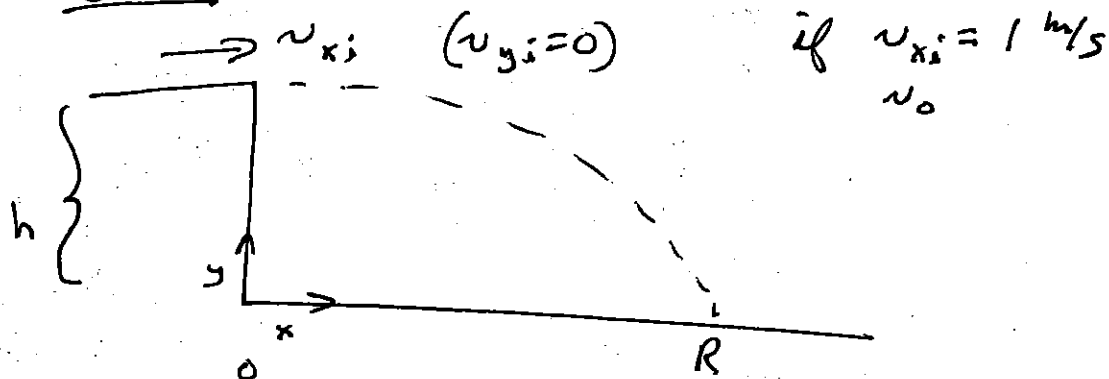
PHYS 1408 CHAPTER 4

2-D Projectile Motion

1/10

This is presented as 3 cases

Case 1



$$x_f = x_i + v_{xi}t + \frac{1}{2}a_x t^2$$

$$x_i = 0 \quad x_f = R \quad a_x = 0$$

$$y_f = y_i + v_{yi}t + \frac{1}{2}a_y t^2$$

$$y_i = h \quad y_f = 0 \quad v_{yi} = 0 \quad a_y = -g$$

$$R = v_{xi}t$$

$$0 = h - \frac{gt^2}{2}$$

$$t = \sqrt{\frac{2h}{g}}$$

$$R = v_0 \sqrt{\frac{2h}{g}}$$

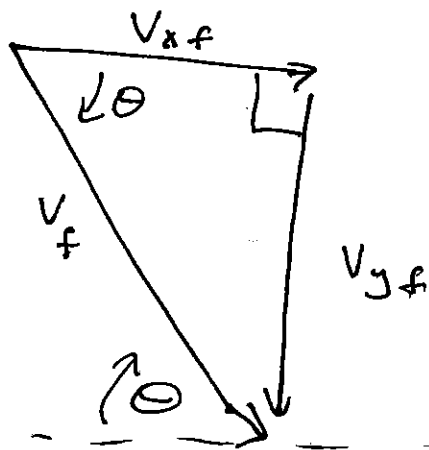
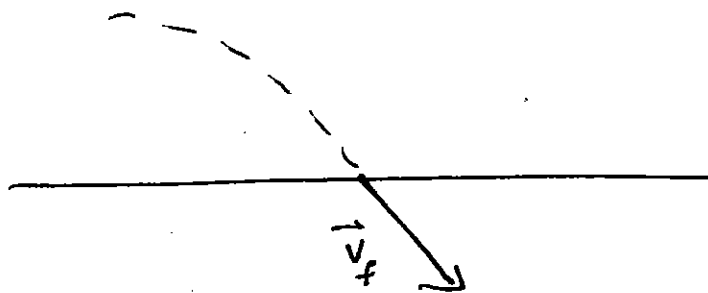
if bar is 0.9 m high & $v_0 = 1 \text{ m/s}$

$$t = \sqrt{\frac{2 \cdot 0.9 \text{ m}}{9.8 \text{ m/s}^2}} \approx 0.43 \text{ s}$$

$$R = v_0 \sqrt{\frac{2h}{g}} \approx 0.43 \text{ m}$$

How does it strike the floor?

2/10



$$\theta = \tan^{-1} \frac{v_{yf}}{v_{xf}}$$

$$v_{yf} = v_{yi} + a_y t$$

$$v_{yi} = 0$$

$$a_y = -g$$

$$v_{yf} = -gt \leftarrow t = \sqrt{\frac{2h}{g}}$$

$$v_{yf} = -\sqrt{2gh} = -4.2 \text{ m/s}$$

$$v_{xf} = v_{xi} = 1 \text{ m/s}$$

$$v_x \text{ is constant } a_x = 0$$

$$\theta = \tan^{-1} \frac{4.2}{1}$$

lose the -, you
are looking at an angle

$$= 76.6^\circ$$

Case 2

kicking a football

3/10



$$V_0 = 20 \text{ m/s}$$

$$\theta = 15^\circ$$

How far does it go?

$$x_i = 0 \quad x_f = R \quad y_i = 0 \quad y_f = 0 \quad V_{x_i} = V_0 \cos \theta \quad V_{y_i} = V_0 \sin \theta$$

$$\sin 15^\circ = 0.259$$

$$\cos 15^\circ = 0.966$$

$$a_y = -g$$

$$a_x = 0$$

$$y_f = y_i + V_{y_i} t + \frac{1}{2} a_y t^2$$

$$0 = 0 + V_0 \sin \theta t - \frac{g}{2} t^2$$

$$V_0 \sin \theta t = \frac{g}{2} t^2$$

$$t = 0 \quad \text{or} \quad V_0 \sin \theta = \frac{g}{2} t$$

$$t = \frac{2 V_0 \sin \theta}{g}$$

$$x_f = x_i + V_{x_i} t + \frac{1}{2} a_x t^2$$

$$R = V_0 \cos \theta t$$

$$R = \frac{2 V_0^2 \sin \theta \cos \theta}{g}$$

Recognize $\sin 2\theta = 2 \sin \theta \cos \theta$

also

$$R = \frac{V_0^2 \sin 2\theta}{g}$$

$$R = \frac{2 v_0^2 \sin \theta \cos \theta}{g}$$

$$= \frac{2 \left(20 \frac{\text{m}}{\text{s}}\right)^2 \sin 15^\circ \cos 15^\circ}{9.8 \text{ m/s}^2}$$

$$= \frac{2 \cdot 20^2 \cdot 0.259 \cdot 0.966}{9.8} \text{ m}$$

$$= 20.4 \text{ m}$$

What is height at highest point?

$$v_{yf} = 0 \quad \text{at top} \quad y_f = h \quad v_i = 0$$

$$v_{yi} = v_0 \sin \theta$$

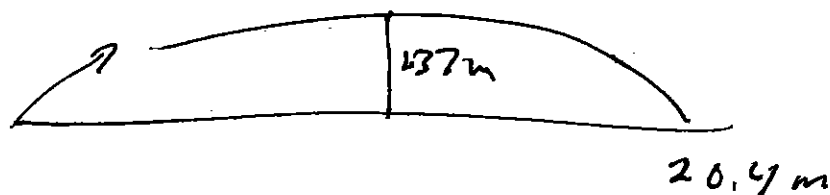
$$v_f^2 = v_i^2 + 2a_y(y_f - y_i)$$

$$0 = v_0^2 \sin^2 \theta - 2g(h - 0)$$

$$2gh = v_0^2 \sin^2 \theta$$

$$\boxed{h = \frac{v_0^2 \sin^2 \theta}{2g}} = \frac{20^2 \cdot 0.259^2}{2 \cdot 9.8}$$

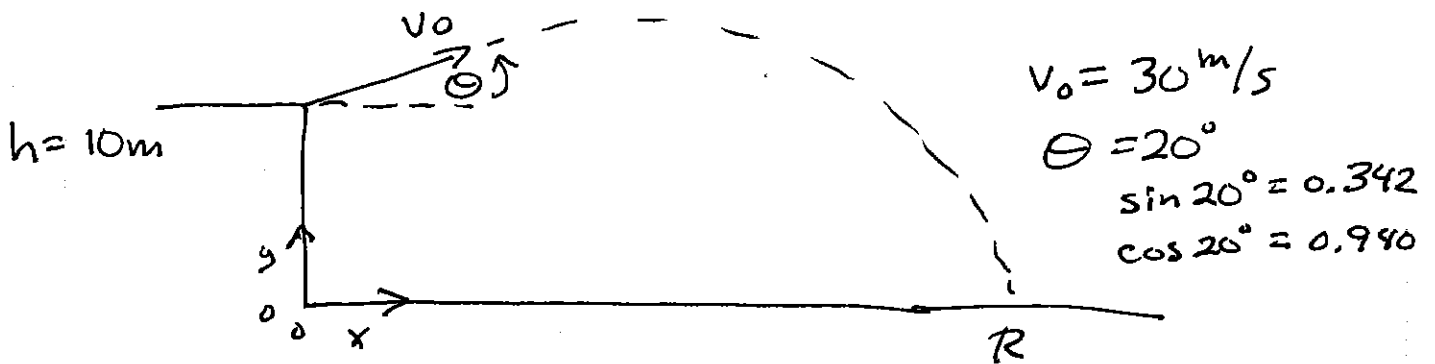
$$h = 1.37 \text{ m}$$



4/10

Case 3 launch from a height

5/10



$$v_{y_i} = v_0 \sin \theta \quad v_{x_i} = v_0 \cos \theta$$

$$y_i = h \quad y_f = 0 \quad x_i = 0 \quad x_f = R \quad a_x = 0 \quad a_y = -g$$

$$x_f = x_i + v_{x_i} t + \frac{1}{2} a_x t^2$$

$$R = 0 + v_0 \cos \theta t + 0$$

$$\boxed{R = v_0 \cos \theta t}$$

Quadratic Formula

$$\boxed{\begin{aligned} ax^2 + bx + c &= 0 \\ x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \end{aligned}}$$

$$y_f = y_i + v_{y_i} t + \frac{1}{2} a_y t^2$$

$$0 = h + v_0 \sin \theta t - \frac{g}{2} t^2$$

$$\underbrace{\frac{g}{2}}_a t^2 - \underbrace{v_0 \sin \theta}_b t + \underbrace{-h}_c = 0$$

$$t = \frac{v_0 \sin \theta \pm \sqrt{v_0^2 \sin^2 \theta - 4 \frac{g}{2} (-h)}}{2 \frac{g}{2}}$$

$$t = \frac{v_0 \sin \theta \pm \sqrt{v_0^2 \sin^2 \theta + 2gh}}{g}$$

$$t = \frac{30 \frac{\text{m}}{\text{s}} 0.342 \pm \sqrt{\left(30 \frac{\text{m}}{\text{s}} 0.342\right)^2 + 2 \cdot 9.8 \frac{\text{m}}{\text{s}^2} 10\text{m}}}{9.8 \text{m/s}^2}$$

6/10

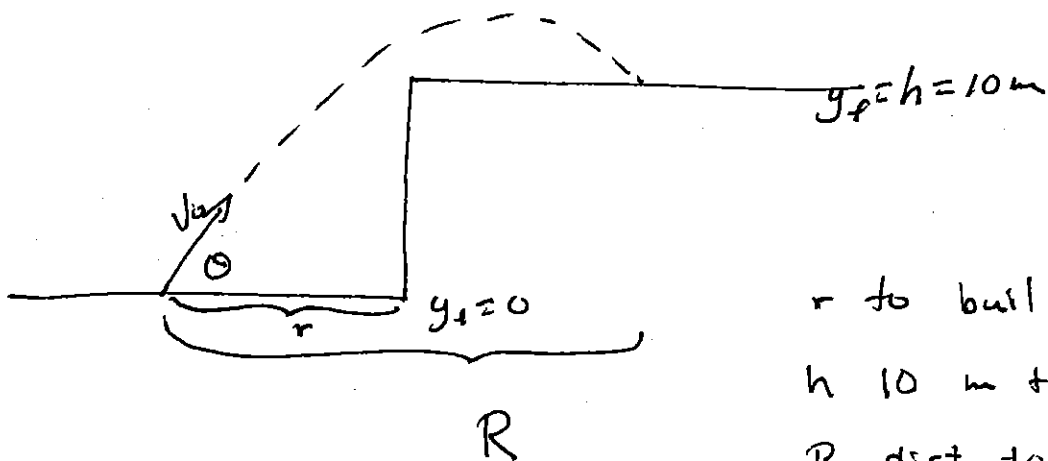
$$t = \frac{10.26 \pm \sqrt{10.26^2 + 196}}{9.8} \text{ s}$$

$$t = \frac{10.26 \pm 17.36}{9.8} \text{ s}$$

$$t = -0.72 \text{ s}, 2.82 \text{ s}$$

$$R = v_0 \cos \theta t = 30 \frac{\text{m}}{\text{s}} 0.940 2.82 \text{ s} = \boxed{79.5 \text{ m}}$$

Case 3 variant



r to building

h 10 m tall

R dist to target

v_0 launch muzzle velocity

$$\theta = 60^\circ$$

$$v_0 = 30 \frac{\text{m}}{\text{s}}$$

$$x_f = x_i + v_{x_i} t + \frac{1}{2} a_x t^2$$

$$x_f = R \quad x_i = 0 \quad v_{x_i} = v_0 \cos \theta \quad a_x = 0$$

$$\boxed{R = v_0 \cos \theta t}$$

$$y_f = y_i + v_{yi} t + \frac{1}{2} a_y t^2$$

7/10

$$h = 0 + v_0 \sin \theta t - \frac{g}{2} t^2 \quad y_f = h \quad y_i = 0 \quad v_{yi} = v_0 \sin \theta$$

$$\frac{g}{2} t^2 - v_0 \sin \theta t + h = 0$$

$$ax^2 + bx + c = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$t = \frac{v_0 \sin \theta \pm \sqrt{v_0^2 \sin^2 \theta - 4 \frac{g}{2} h}}{g}$$

$$t = \frac{v_0 \sin \theta \pm \sqrt{v_0^2 \sin^2 \theta - 2gh}}{g}$$

$$t = \frac{30 \frac{m}{s} 0.866 \pm \sqrt{(30 \frac{m}{s} 0.866)^2 - 2 \cdot 9.8 \frac{m}{s^2} 10m}}{9.8 \frac{m}{s^2}}$$

$$t = \frac{25.98 \pm \sqrt{674.96 - 196}}{9.8}$$

$$t = \frac{25.98 \pm 21.89}{9.8}$$

$$t = \underset{\text{root}}{0.417 s} \quad , \quad \underset{\text{root}}{4.88 s}$$

$$R_+ = v_0 \cos \theta t_+ = 30 \frac{m}{s} 0.5 \cdot 4.88 s = 73.2 m$$

$$R_- = v_0 \cos \theta t_- = 30 \frac{m}{s} 0.5 \cdot 0.417 s = 6.26 m$$

if building close to mortar - hits at 6.26 m

if $V_0 = 15 \text{ m/s}$ $h = 20 \text{ m}$ $\theta = 30^\circ$

slower ~~lower~~ higher lower

8/10

$$t = \frac{V_0 \sin \theta \pm \sqrt{V_0^2 \sin^2 \theta - 2gh}}{g}$$

$$= \frac{15 \cdot 0.5 \pm \sqrt{(15 \cdot 0.5)^2 - 2 \cdot 9.8 \cdot 20}}{9.8}$$

$$= \frac{7.5 \pm \sqrt{7.5^2 - 392}}{9.8}$$

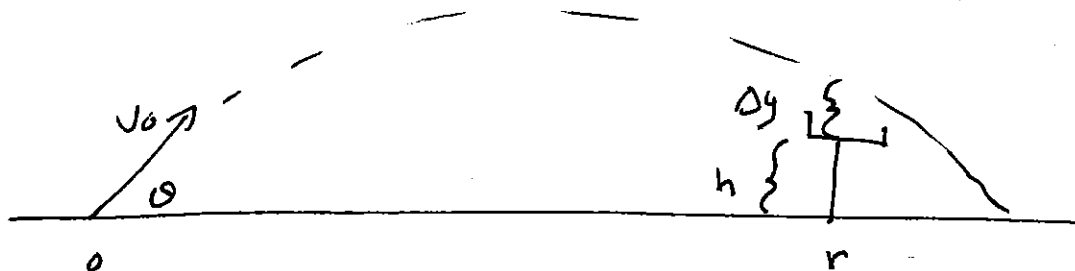
$$= \frac{7.5 \pm \sqrt{-336}}{9.8} \quad \text{s imaginary times!}$$

never gets as high as building.

Field Goal Problem

9/10

Assisting in 56 42 victory over
U of Lx



A field goal is kicked at 25 m/s at 30°

at a 3m goal 35m away

$r = 35 \text{ m}$ $h = 3 \text{ m}$ $\theta = 30^\circ$ $v_0 = 25$ $y = ?$ $\Delta y = ?$

$$x = ? \quad x_f = x_i + v_{x_i} t + \frac{1}{2} a_x t^2$$

$$r = 0 + v_0 \cos \theta t + 0$$

$$r = v_0 \cos \theta t$$

$$t = \frac{r}{v_0 \cos \theta} = \frac{35 \text{ m}}{25 \frac{\text{m}}{\text{s}} \cdot 0.866} \approx 1.617 \text{ s}$$

$$y = ? \quad y = y_i + v_{y_i} t + \frac{1}{2} a_y t^2$$

$$y = 0 + v_0 \sin \theta t - \frac{g}{2} t^2$$

$$y = 25 \frac{\text{m}}{\text{s}} \cdot 0.5 \cdot 1.617 \text{ s} - \frac{9.8 \frac{\text{m}}{\text{s}^2}}{2} (1.617 \text{ s})^2$$

$$y = 20.21 - 12.81 = 7.4 \text{ m}$$

$$\Delta y = y - h = 4.4 \text{ m}$$

Change 30° kick to 28°

10/10

$$t = \frac{r}{v_0 \cos \theta} = \frac{35}{25 \cdot 0.8829} = 1.5865$$

$$y = 25 \cdot 0.4695 \cdot 1.586 - \frac{9.8}{2} \cdot 1.586^2$$

$$= 18.61 - 12.32$$

$$y = 6.29 \text{ m}$$

$$\Delta y = 3.29 \text{ m}$$